

Images & objectivity

Photography, data visualization, and the view from nowhere

What does it mean for something to be
“objective” or “objectively true”?

Accountability, rationality, reproducibility, falsifiability...

Objective judgment

- **Judgment in what gets counted:** calculating a GDP with paid nanny vs. stay-at-home parent
- **Judgment in the categories:** US census changes racial categories every decade; Diagnosis and Statistical Manual of Mental Disorders (DSM-V)
- **Judgment built into the instrument:** Shirley cards, pulse oximeters, crash-test dummies
- **Judgment in presentation:** map distortions
- **Objectivity by agreement:** standard time zones and railroads, robot umpires

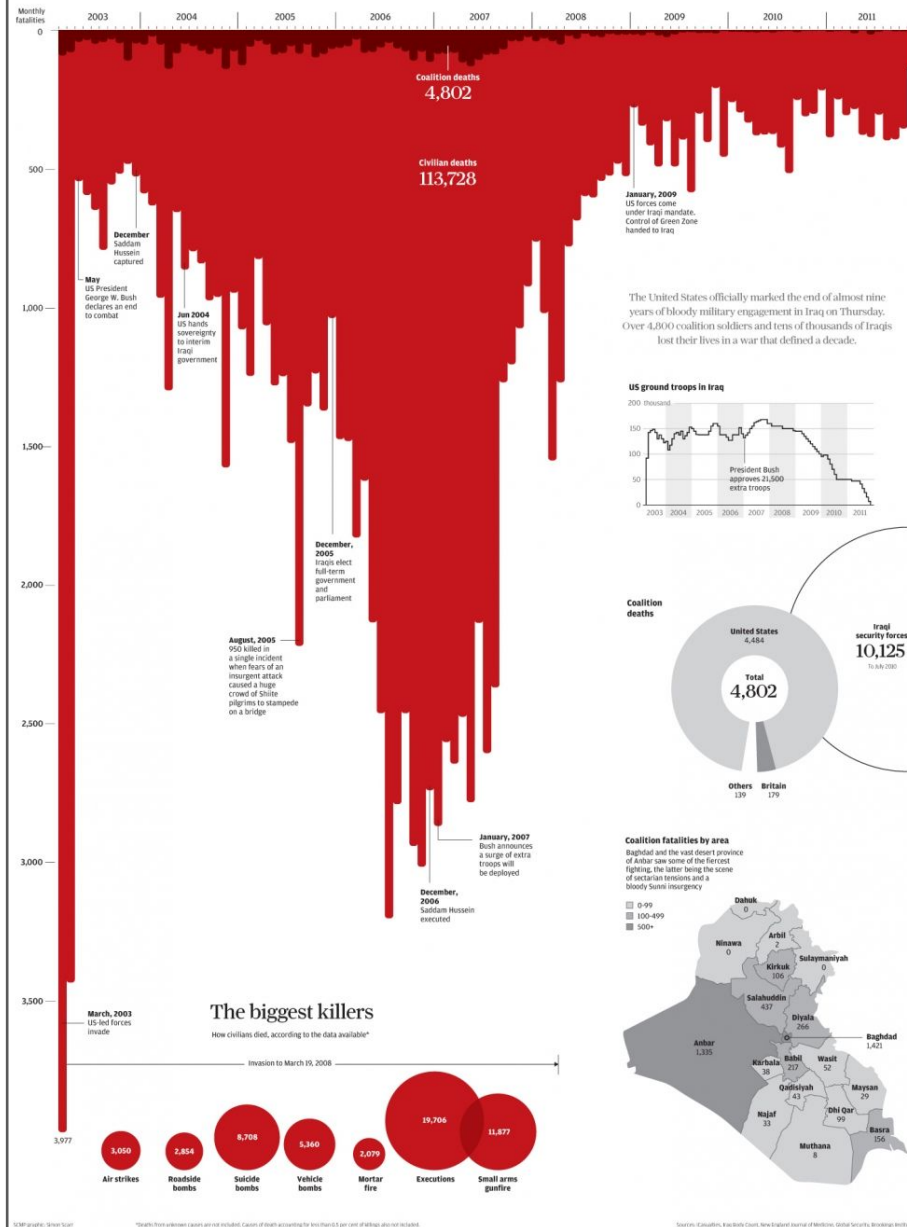




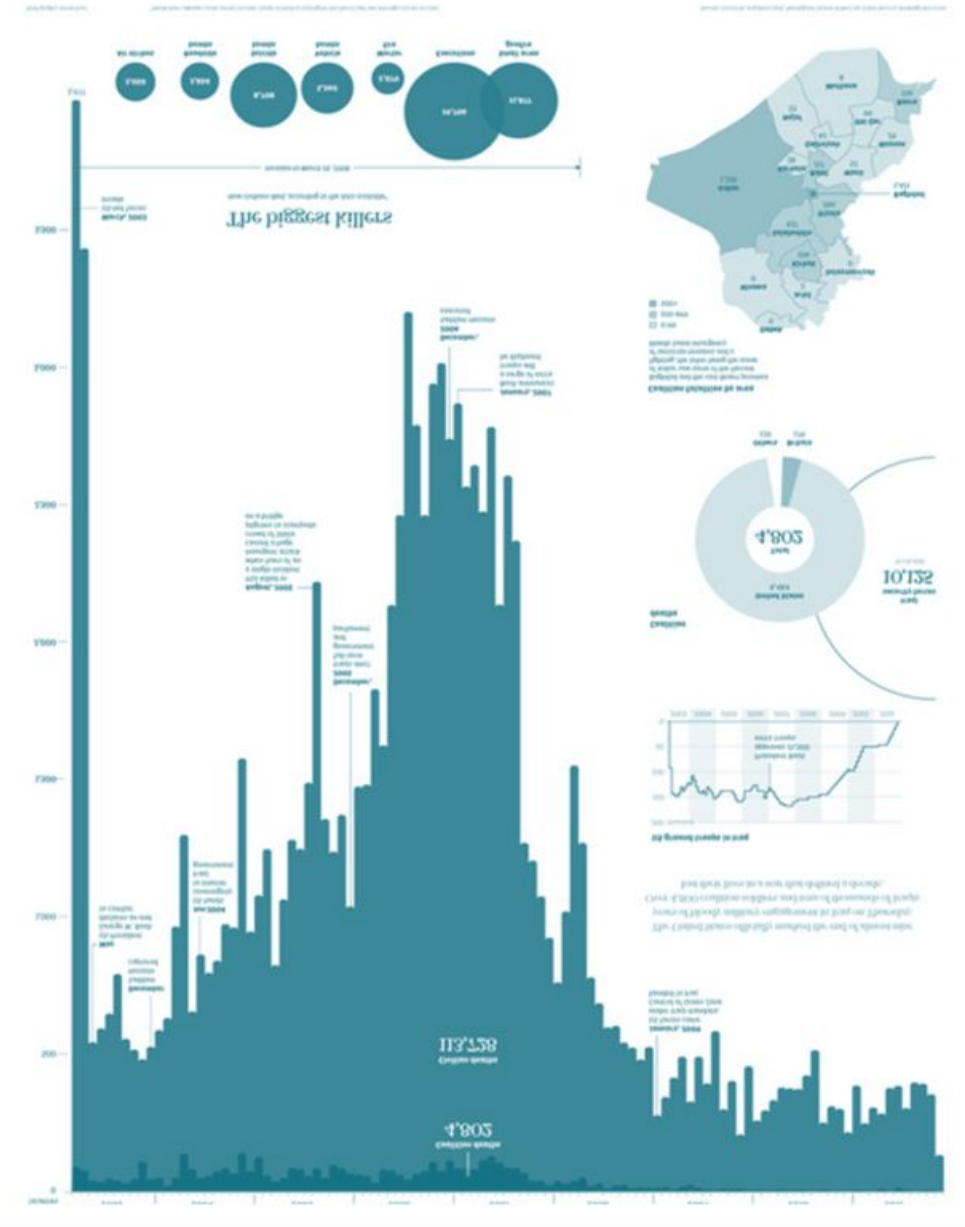




Iraq's bloody toll



Iraq: Deaths on the decline



medium matters!

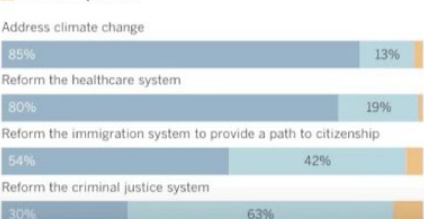
Climate becomes voters' top issue

[Priorities, from A1] changes to the criminal justice system aimed at reducing the number of people in prison.

Overall, 47% chose climate change, followed by 32% picking healthcare reform, 14% immigration and 3% criminal justice.

"For the more than 40 years I've been polling Californians, protecting the environment has always been a high priority to voters here. But now, with the negative consequences of global warming appearing to bear down hard on the state, con-

Climate, healthcare the priorities for likely primary voters



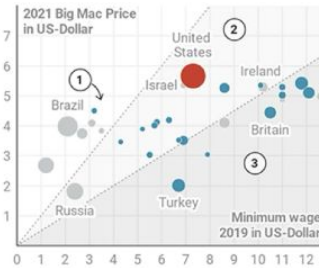
college degrees. His supporters were slightly more likely to put reforming the healthcare system at the top of their list.

Biden's supporters were also more likely than supporters of other candidates to list immigration as the No. 1 priority.

Sen. Bernie Sanders of Vermont has made reform of "Medicare for all" a centerpiece of his campaign. His supporters were what more likely to pick climate change as the top priority.

How long do people need to work at minimum wage to buy one Big Mac?

2021 Big Mac price in US-Dollar according to the Economist Big Mac Index and 2019 minimum wage in US-Dollar in selected countries. The bigger a country circle, the bigger the population. European countries are highlighted in blue.

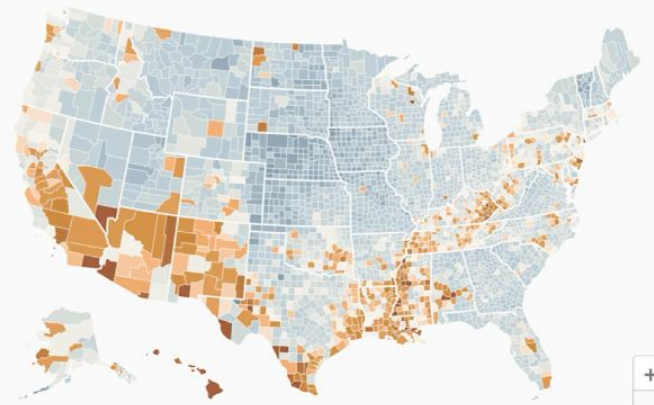


- 1 People in these countries need to work more than 1h at minimum wage to buy one Big Mac.
- 2 ...need to work between 30min and 1h for one Big Mac.
- 3 ...need to work less than 30min for one Big Mac.

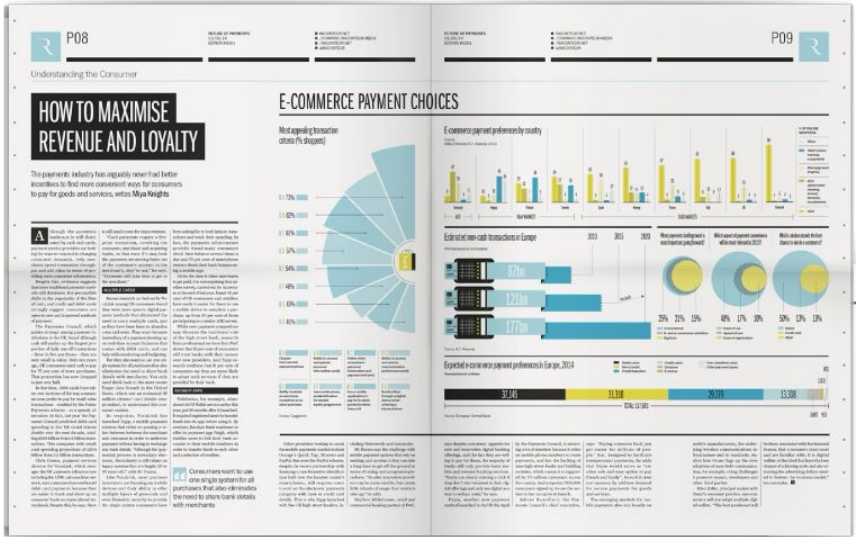
Based on a chart by RJ Andrews.
Source: The Economist (Big Mac prices) and OECD (minimum wages)
• Created with Datawrapper

Unemployment rate, October 2020

Counties with a unemployment rate above the national average (6.9%) and below in October 2020

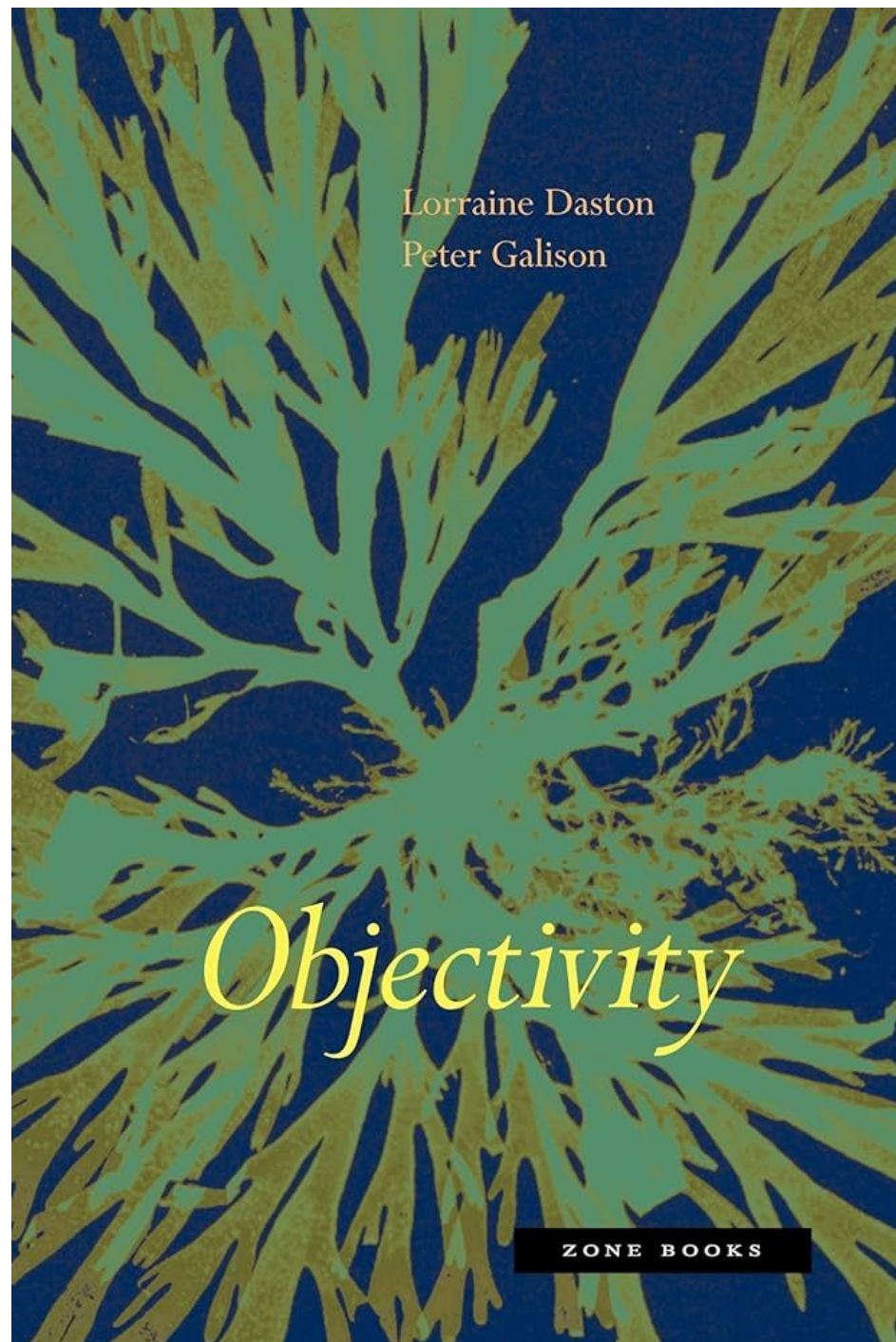


Source: BLS/LAUS



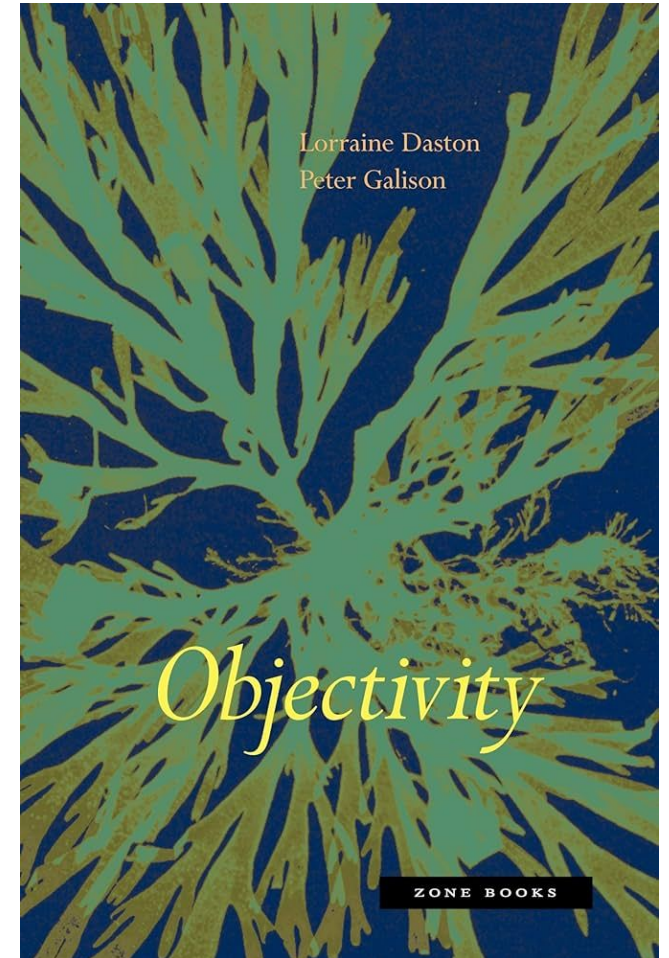
Discussion questions

1. What does each image show?
2. Is any one of them false? What would it take to prove that?
3. What decisions did the maker make? List every one you can find
4. Where is the viewer standing? Who could actually see this?
5. What would you need to know before trusting any of them?



History of objectivity

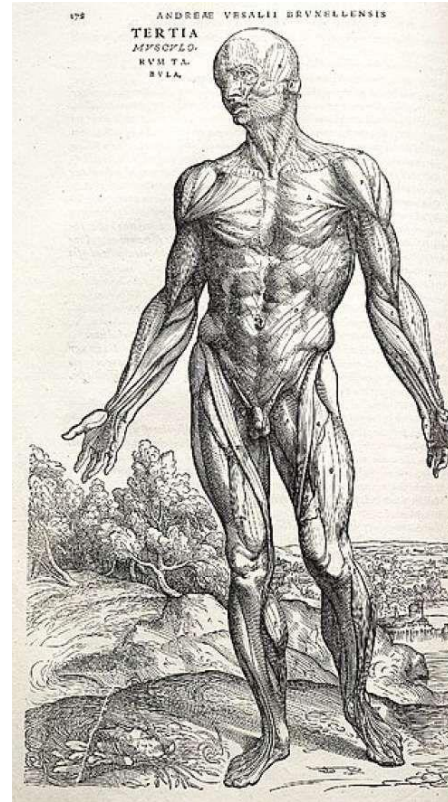
- Truth to nature
- Objectivity: structural and mechanical
 - Trained judgment & virtual witnessing
- Trained judgment



TRUTH TO NATURE

Truth-to-nature

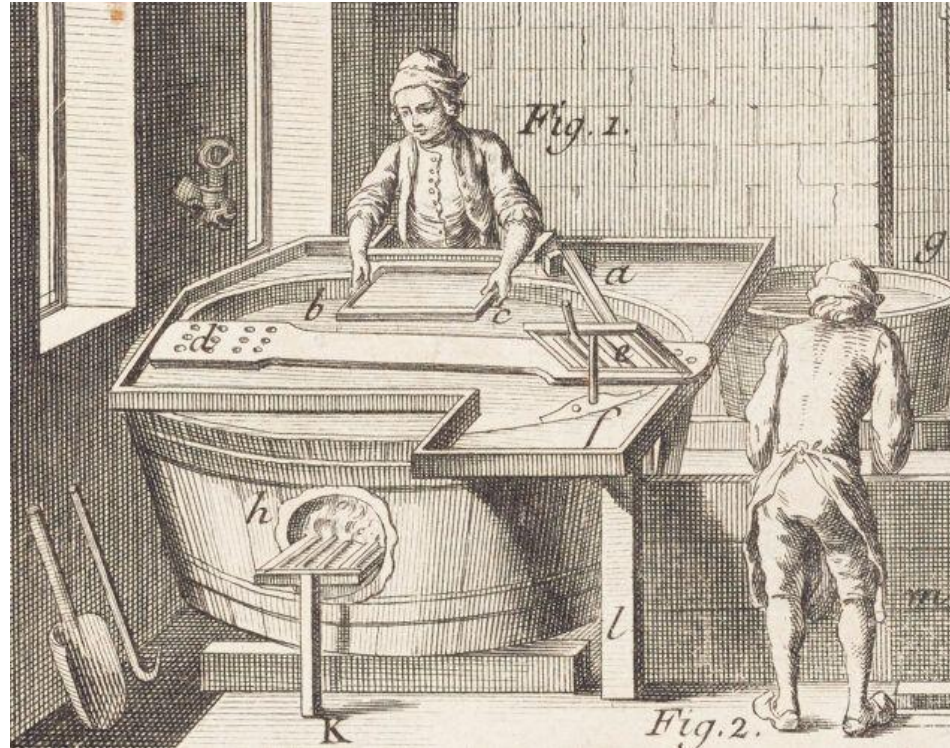
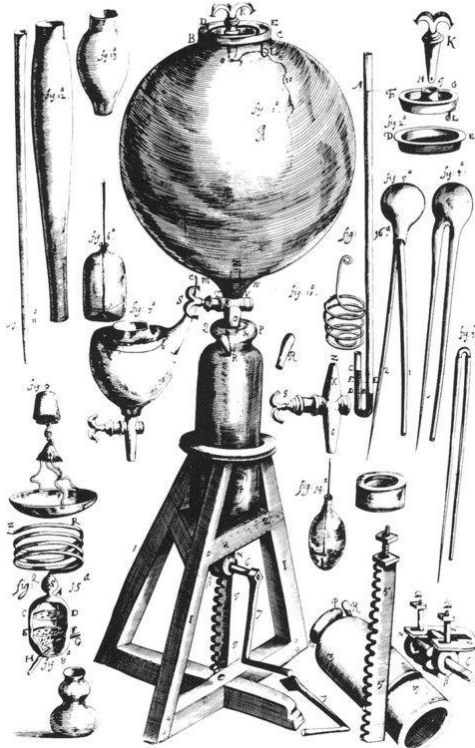
- Select best, most “representative” specimens
- *Composite*, idealized image
 - Anatomical (Vesalius) illustrations
 - Botanical / zoological illustrations (Haeckel)



Models of nature



Virtual witnessing



- Detailed engravings let readers “see” an experiment or a workshop they never visited
- More witnesses, more credible facts
- Boyle’s Air Pump, papermaking in *Encyclopédie*

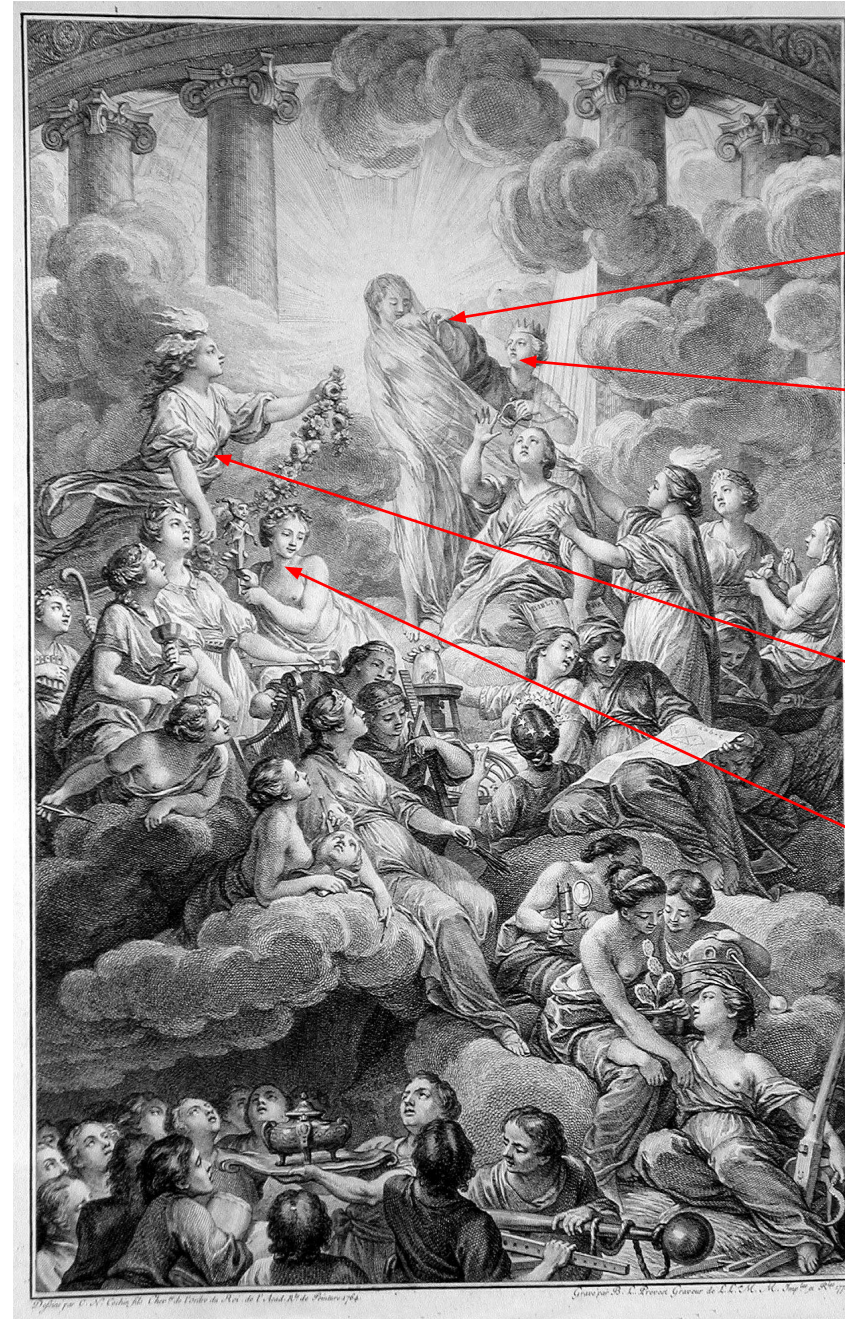
***Encyclopédie* frontispiece**

“We see on top Truth between Reason and Imagination;
Reason is trying to snatch off her veil, Imagination is preparing to embellish her;
below this group, a crowd of speculative philosophers; lower down, the troupe of artists...”



***Encyclopédie* frontispiece**

“We see on top Truth between Reason and Imagination; Reason is trying to snatch off her veil, **Imagination is preparing to embellish her**; below this group, a crowd of speculative philosophers; lower down, the troupe of artists...”



Truth

Reason

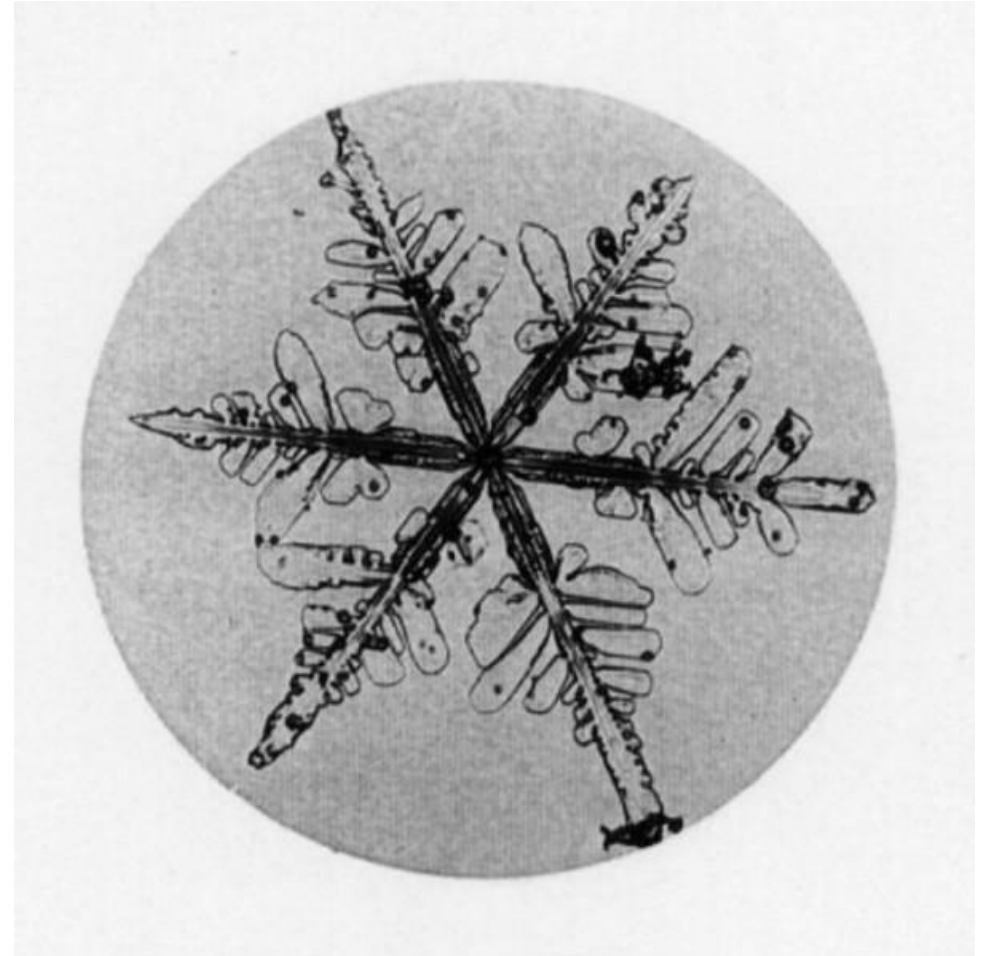
Imagination

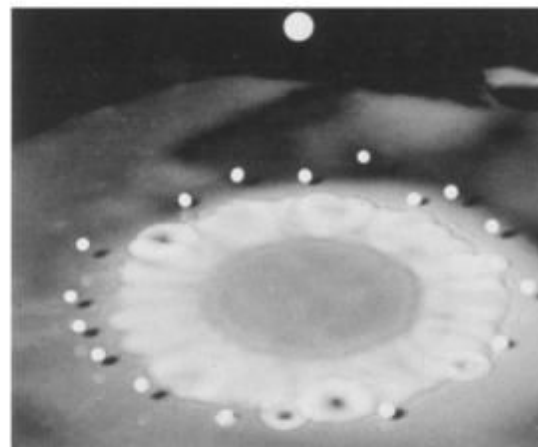
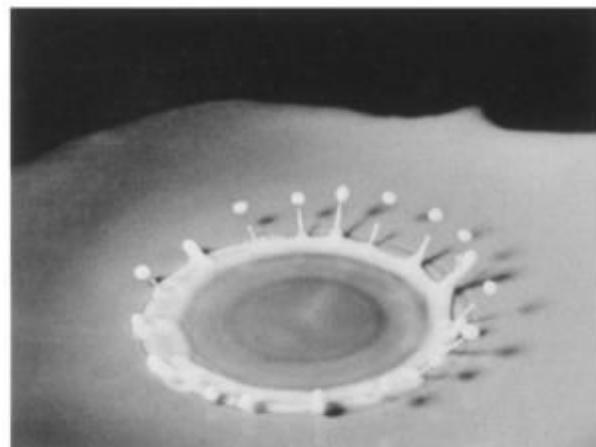
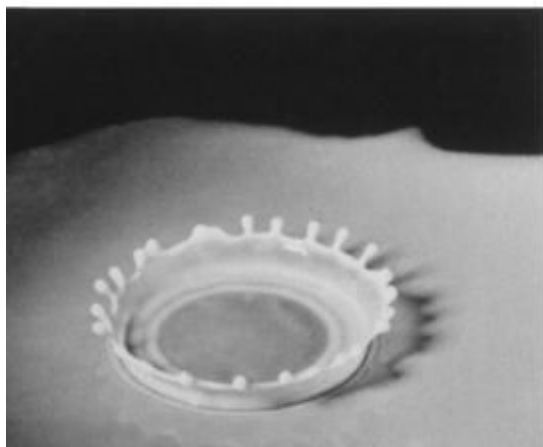
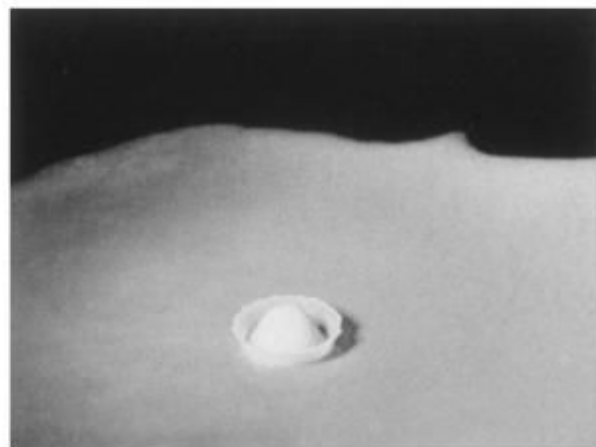
Theology

MECHANICAL OBJECTIVITY

Mechanical objectivity

- Mechanical reproduction of reality (e.g. photographs and microscopes)
- **Many** copies of imperfect specimens are better than one that conflates many exemplars
 - Gustav Hellmann, with microphotographs by Richard Neuhauss, Schneekrystalle (1893).







Editing and partial perspective

- How do edits change what and how you see a film?
- How do “invisible” edits give you a partial perspective?
- Partial perspective and datasets

The Values Encoded in Machine Learning Research

Abeba Birhane*
abeba@mozillafoundation.org
Mozilla Foundation & School of
Computer Science, University College
Dublin
Dublin, Ireland

William Agnew*
wagnew3@cs.washington.edu
Paul G. Allen School of Computer
Science and Engineering, University
of Washington
Seattle, USA

Pratyusha Kalluri*
pkalluri@stanford.edu
Computer Science Department,
Stanford University
Palo Alto, USA

Ravit Dotan*
ravit.dotan@berkeley.edu
Center for Philosophy of Science,
University of Pittsburgh
Pittsburgh, USA

Dallas Card*
dale@umich.edu
School of Information, University of
Michigan
Ann Arbor, USA

Michelle Bao*
baom@stanford.edu
Computer Science Department,
Stanford University
Palo Alto, USA

ABSTRACT

Machine learning currently exerts an outsized influence on the world, increasingly affecting institutional practices and impacted communities. It is therefore critical that we question vague conceptions of the field as value-neutral or universally beneficial, and investigate what specific values the field is advancing. In this paper, we first introduce a method and annotation scheme for studying the values encoded in documents such as research papers. Applying the scheme, we analyze 100 highly cited machine learning papers published at premier machine learning conferences, ICML and NeurIPS. We annotate key features of papers which reveal their values: their justification for their choice of project, which attributes of their project they uplift, their consideration of potential negative consequences, and their institutional affiliations and funding sources. We find that few of the papers justify how their project connects to a societal need (15%) and far fewer discuss negative potential (1%). Through line-by-line content analysis, we identify 59 values that are uplifted in ML research, and, of these, we find that the papers most frequently justify and assess themselves based on Performance, Generalization, Quantitative evidence, Efficiency, Building on past work, and Novelty. We present extensive textual evidence and identify key themes in the definitions and operationalization of these values. Notably, we find systematic textual evidence that these top values are being defined and applied with assumptions and implications generally supporting the centralization of power. Finally, we find increasingly close ties between these highly cited papers and tech companies and elite universities.

*All authors contributed equally to this research.



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KEYWORDS

Encoded values of ML, ICML, NeurIPS, Corporate ties, Power asymmetries

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1 INTRODUCTION

Over recent decades, machine learning (ML) has risen from a relatively obscure research area to an extremely influential discipline, actively being deployed in myriad applications and contexts around the world. Current discussions of ML frequently follow a historical strain of thinking which has tended to frame technology as “neutral”, based on the notion that new technologies can be unpredictably applied for both beneficial and harmful purposes [64]. This claim of neutrality frequently serves as an insulation from critiques of AI and as permission to emphasize the benefits of AI [47, 58, 63], often without any acknowledgment that benefits and harms are distributed unevenly. Although it is rare to see anyone explicitly argue in print that ML is neutral, related ideas are part of contemporary conversation, including these canonical claims: long term impacts are too difficult to predict; sociological impacts are outside the expertise or purview of ML researchers [28]; critiques of AI are really misdirected critiques of those deploying AI with bad data (“garbage in, garbage out”), again outside the purview of many AI researchers; and proposals such as broader impact statements represent merely a “bureaucratic constraint” [3]. ML research is often cast as value-neutral and emphasis is placed on positive applications or potentials. Yet, the objectives and values of ML research are influenced by many social forces that shape factors including what research gets done and who benefits.¹ Therefore, it is important to challenge perceptions of neutrality and universal benefit, and document and understand the emergent values of the

¹For example, ML research is influenced by social factors including the personal preferences of researchers and reviewers, other work in science and engineering, the interests of academic institutions, funding agencies and companies, and larger systemic pressures, including systems of oppression.



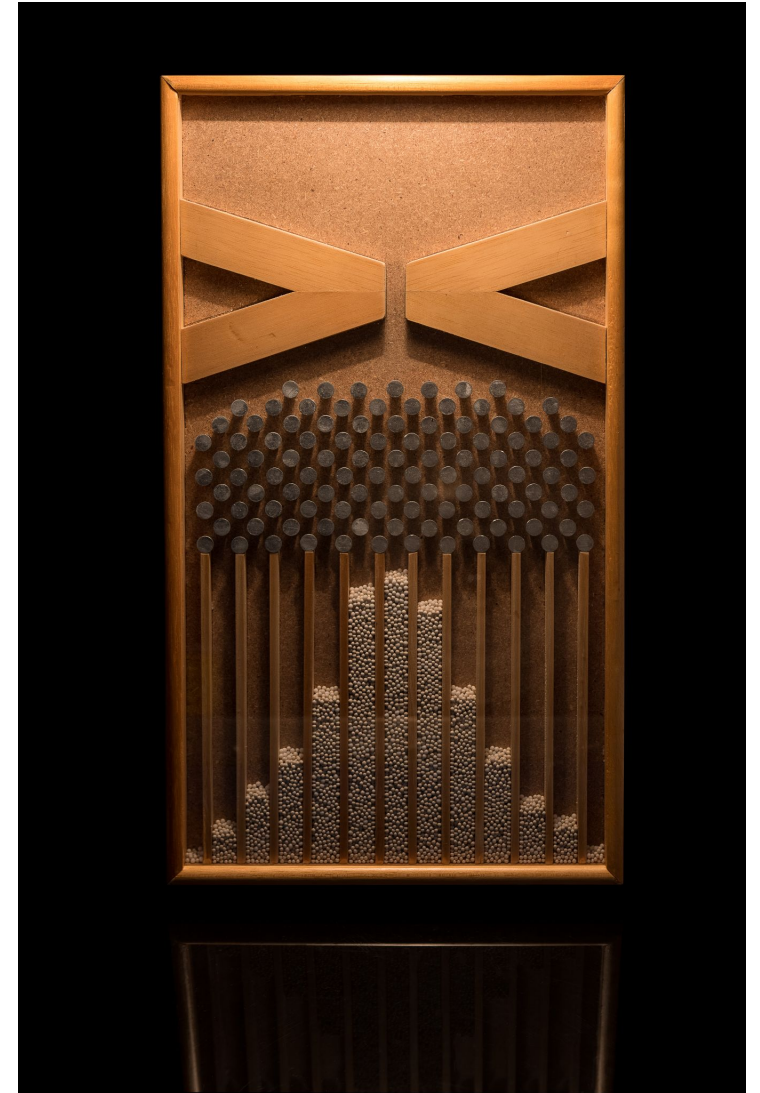
Vox

NORMAL

STRUCTURAL OBJECTIVITY

Structural objectivity

- Mathematical laws, logical relationships among members of a group or organism, about *systems*
- Based on systematic analysis and abstraction
 - Galton Board and normal distribution, Poincaré's disk model



TRAINED JUDGMENT

Trained judgment

- Raw data is smoothed by an expert to study it better
- Unsmoothed and smoothed data are presented to the judgment of the user or reader



Stages in the history of objectivity

Epistemic virtues	When	The ideal
Truth-to-nature	18th–early 19th c.	Show the ideal type behind the specimen; the naturalist corrects the artist
Mechanical objectivity	Mid-19th c. onward	Let nature speak for itself; restrain the observer
Structural objectivity	Late 19th–early 20th c.	Objectivity lies in invariant relations, not images
Trained judgment	20th c.	Expert eyes interpret and classify patterns

Situated knowledges (Donna Haraway)

- Can knowledge be objective and accountable to where it comes from?
- All knowledge comes from positional perspectives: partial, located, and accountable
- ***This does not mean that scientific truth and consensus do not exist!***

An aerial photograph of an oil field. Numerous drilling rigs are visible across a flat, grassy landscape. A dirt road or pipeline runs diagonally from the bottom left towards the center. A small dark vehicle is on the road. The text "Where Do You Put the Camera?" is overlaid in white, with "the Camera?" in italics.

Where Do You Put
the Camera?





WIRED